An Information Entropy Feature Preservation-Based Band Selection and Noise Reduction Method for Hyperspectral Image Classification

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***Abstract*—Effectively reducing spectral redundancy while preserving the distinguishability of land cover is key to hyperspectral image classification (HSIC). Moreover, the presence of noise also interferes with the discriminability of spectral features. To address these issues, this article introduces a novel HSIC method that integrates band selection (BS) and discriminative feature extraction to achieve high performance of land-cover classification. First, by incorporating the concept of information entropy, the information entropy feature (IEF) is designed to describe the relationship between spectral bands, and then, the representative band group is selected based on the designed IEF. Second, a novel tensor subspace representation (TSR) model is proposed to separate the land cover of interest and noise component for the selected band group. Next, in light of different shapes and sizes exhibited by different land-cover, multiscale spatial–spectral features (MSFs) are used as supplementary information. Finally, the extracted class feature is used to guide and refine the extraction of land cover, further optimizing the classification performance. The results on multiple datasets show that the proposed method outperforms existing methods.**

***Index Terms*—Classification, dimensionality reduction (DR), feature fusion, hyperspectral image (HSI), tensor subspace representation (TSR).**

I. Introduction

AS AN important tool in remote sensing, hyperspectral imaging technology can capture information on the reflectance or radiance of objects across various wavelengths, thereby producing images with rich spectral details. It finds widespread application in fields such as anomaly detection [1], [2] and hyperspectral image classification (HSIC) [3]. Among them, HSIC aims to assign a semantic label to each pixel in an image. It holds significant practical value in agriculture, environmental monitoring, and urban planning [4]. Various methods, including both classic approaches (e.g., support vector machines [5] and multinomial logistic regression [6]) and deep learning techniques [7], have been proposed for this task. While many of these methods achieve promising results, noise and redundancy in HSIs remain challenges to improving classification performance.

In real-world scenarios, HSIs are often corrupted by noise, leading to degraded image quality and lower classification accuracy. Numerous HSI denoising algorithms have been proposed for preprocessing before classification to address this issue. These methods can generally be categorized into low-rank (LR) matrix approximation, sparse representation, total variation (TV) model, nonlocal means, LR tensor [8], [9], [10], [11], and other data-driven approaches [12]. Although these techniques are intended to recover high-quality HSI, they do not incorporate task-specific considerations during data restoration. As a result, the restored data may not be well-suited for HSIC tasks. In response to this challenge, some research has focused on classifying noisy HSI. For example, Zhong and Wang [13] proposed a joint learning method to address the issue of separately handling noise removal and classification, allowing denoising and classification to benefit from each other. Similarly, as the weak correlation between denoising and high-level vision tasks in traditional image processing, Liu et al. [14] systematically studied the mutual influence between low-level image denoising and high-level vision networks, such as image classification and semantic segmentation. Yang et al. [15] proposed a discriminative LR representation model and, based on this, developed a superpixel-based induced classification framework. In this framework, superpixel segmentation is iteratively refined using image restoration results, and the segmentation outcomes facilitate the recovery of more discriminative HSI. Li et al. [16] defined a compound loss function to enable end-to-end learning, allowing denoising and classification to mutually benefit from each other. However, HSI contains hundreds of bands, making full-band processing computationally intensive. The restored high-dimensional data often retain excess spectral information that is unnecessary for classification tasks,

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much of which is redundant due to the correlations between bands. The increase in feature dimensions necessitates a large amount of training samples, which may result in the Hughes phenomenon, where classification models struggle to achieve satisfactory performance.

To enhance classification performance and reduce the computational load in subsequent processing, some research has focused on processing data after dimensionality reduction (DR) [17]. For example, Kang et al. [18] decomposed dimension-reduced images into reflectance and shadow components based on independent and identically distributed components. This approach enhances classification accuracy by eliminating irrelevant spatial information, such as shadows and noise. Similarly, Cao et al. [19] employed multiple spatial feature extraction techniques to capture complementary information from selected band subsets. Among these, the wavelet transform was used to preserve the majority of information from the original bands while effectively suppressing noise. Li et al. [20] demonstrated that applying segmented principal component analysis (SPCA) and Gaussian pyramids for feature extraction and fusion can effectively reduce noise, thereby significantly improving land-cover classification performance. To reduce the impact of noise in images, Kang et al. [21] employ a band averaging and PCA-based edge-preserving filtering (EPF) to improve land-cover classification. Fu et al. [22] proposed an unsupervised DR framework with feature interpretability by integrating band selection (BS) and spatial denoising methods, to extract low-dimensional spectral–spatial features from HSI. While these methods have shown promising progress, several challenges remain. First, the new features generated through various linear or nonlinear transformation techniques for DR often lead to the loss of their intuitive physical meaning. Second, these methods typically do not directly consider the 3-D structural features of HSI, which may result in the loss of important structural information and impact classification performance.

With the development of deep learning, convolutional neural networks (CNNs), autoencoders (AEs), and their variants have been widely applied to feature extraction and classification tasks in HSI [23], [24], [25]. For example, some studies use 3D-CNN to directly process the 3-D structure of HSI to automatically extract spatial–spectral joint features [26]. Other works employ stacked AEs (SAEs) or transformers to model long-range dependencies and compress redundant information [27]. Although these methods exhibit excellent classification performance, they impose high demands on the computational resources.

Motivated by the aforementioned observations, we propose an information entropy feature (IEF) preservation-based BS and noise reduction (IEFPNR) method for HSIC. IEFPNR achieves interpretable feature extraction while interference suppression through a physically meaningful BS strategy and data decomposition. The framework is illustrated in Fig. 1. Specifically, this method comprises two main components: BS and feature extraction. In BS, the IEF is employed to select bands in HSI that contribute significant incremental information, thereby enhancing the representativeness and interpretability of the selected band subset. Regarding feature extraction, a tensor subspace representation (TSR) method is proposed to extract features from the selected band groups while simultaneously resisting noise interference. Additionally, considering that ground objects exhibit diverse spatial structural characteristics, a multiscale spatial–spectral feature (MSF) extraction module is integrated. This module supplements the multiscale information of the extracted ground objects to construct enhanced and robust features. These enhanced features effectively depict ground object information with increased resistance to interference. Finally, to suppress intraclass spectral variability and enhance intraclass similarity, the preclassification result is used to extract spatial relationships. This information is then used to guide the update process of feature extraction, ultimately leading to further improvement in classification accuracy. The contributions of this article can be summarized as follows.

1) We propose a DR method for HSIC, namely, IEFPNR. By integrating BS and discriminative feature extraction, IEFPNR can effectively capture multiscale, low-spectral-redundancy, and interference-robust discriminative features of objects.
2) We propose an IEF-preserving BS (IEFPBS) method. By constructing an IEF curve to characterize the spectral information structure across bands, IEFPBS can select the band subset with high discriminability and low redundancy.
3) We develop an improved TSR model that utilizes weighted tensor nuclear norm (TNN) and tensor decomposition tensor singular value decomposition (t-SVD) to directly learn the LR properties of the coefficient tensor from HSI. Additionally, MSF is incorporated into IEFPNR to adapt to the model's need for representing diverse structures of objects.
4) We propose a feature enhancement strategy that leverages preclassification to obtain class information and spatial relationships, thereby enhancing the representation capability of TSR and MSF for object features. The experimental results validate the effectiveness and robustness of the proposed method.

II. Proposed Method

A. IEFPBS

IEFPBS combines information theory with curvature, aiming to exploit the unique relationship among hundreds of bands for DR and improving land-cover classification accuracy. IEFPBS is mainly based on the following two criteria: 1) construct the IEF curve. These extrema reflect abrupt changes in spectral band information, such as transition zones between redundant and complementary bands and 2) determine the number of selected bands n_{BS}. Then, the spectral bands are divided into n_{BS} intervals. Within each interval, the band with the maximum curvature in the IEF curve is selected to ensure spectral diversity.

In this article, the virtual dimensionality (VD) value [28], [29] is used to determine the number of selected bands n_{BS}. Specifically, n_{BS} is set as the larger value obtained from two VD estimation methods [30]: Harsanyi–Farrand–Chang (HFC), denoted as $\mathrm{VD}^{\mathrm{HFC}}$, and noise-whitened HFC, denoted as $\mathrm{VD}^{\mathrm{NWHFC}}$.

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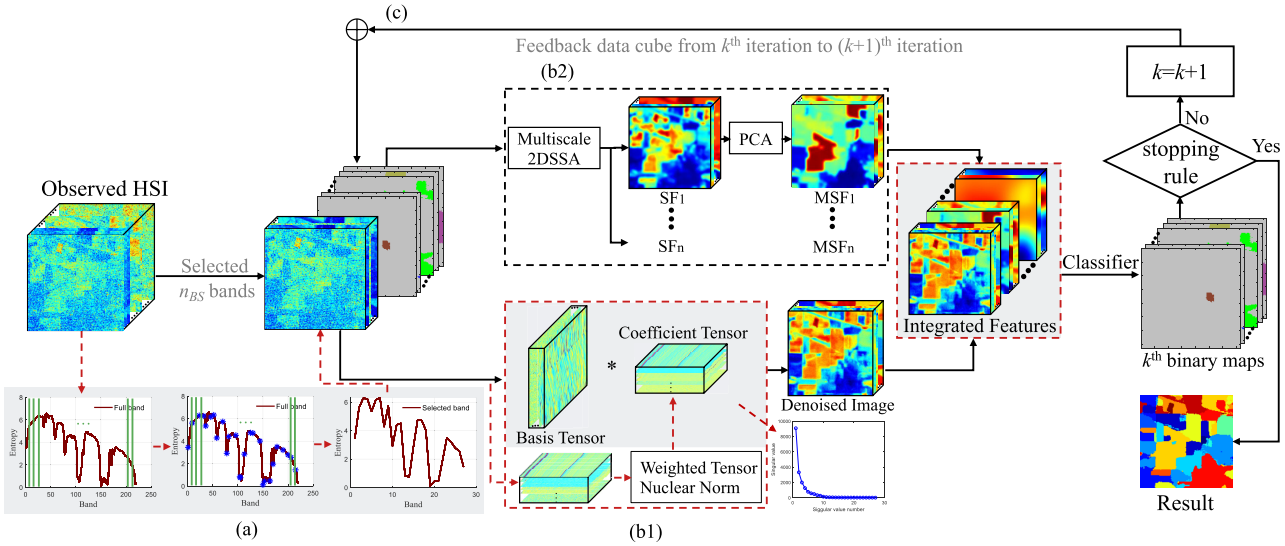


Fig. 1. Flowchart of the proposed IEFPNR method for HSIC. (a) IEFP-based BS. (b1) TSR feature extraction. (b2) Multiscale feature extraction. (c) Class feature guidance.

Due to the strong correlation between spectral bands, the HSI contains significant redundant information. To address this issue of informational redundancy, Chang et al. [31] point out that since entropy is an infinite-order statistical criterion, it generally performs better than second-order statistical criteria in measuring information redundancy. In particular, information theory is likely one of the most effective tools for providing theoretical support [32] and has garnered increasing attention in recent years, for example, entropy-based band prioritization methods [33], sample entropy and class entropy mentioned in [34], as well as the self-information-based class information for BS in HSIC [35]. Recently, a BS method using Shannon's second coding theorem was proposed, which finds prominent peaks in the spectral mutual information (SMI)-generated curve to select the optimal band set [36]. These studies explore the application of information theory in BS and HSIC and demonstrate its effectiveness; However, there is still room for further research. For instance, BS methods relying solely on prominent peaks often overlook interband relationships and dynamic changes in information, resulting in a loss of discriminative features. In this article, we integrate information theory and curvature analysis to present an IEF curve. This curve uses entropy to reflect changes in information across different spectral bands, while curvature is introduced as an effective tool [37] to capture variations of information between bands and identify significant spectral bands. The value of information entropy is calculated as

$$H(b) = -\sum_{w\in\phi} p(w)\log p(w) \tag{1}$$

where w is the value of grayscale color, and ϕ denotes the set of all grayscale values on band b. The curvature of IEF curve can be calculated by the following equation:

$$k(b) = \frac{|H''(b)|}{\left(1+(H'(b))^2\right)^{3/2}} \tag{2}$$

where H represents the IEF curve constructed by all spectral bands, $H'(b)$ is the first derivative of the curve at band b, $H''(b)$ is the second derivative, and $k(b)$ represents the curvature of the curve at a given band b.

Unlike BS methods [38], [39], which focus on selecting spectral bands with high entropy, IEFPBS aims to achieve BS by preserving the informational characteristics between bands, thereby maintaining the richness and discriminative power of spectral information. Moreover, by including various information about spectral bands, IEFPBS enhances the diversity and comprehensiveness of the selected bands. Taking the Indian Pines (IP) scene as an example, the IEF curve of the full band is shown in Fig. 2(a). The IEF curve is divided into n_{BS} band intervals, and the curvature of each band interval is calculated to characterize the spectral information variation features. Furthermore, by selecting bands with significant information variability, the land-cover discrimination capability of the data cube is preserved while reducing interference from redundant information. Fig. 2(b) presents the IEF curve for the band subset selected by IEFPBS. The results demonstrate that the feature curve in Fig. 2(b) has characteristics similar to those in Fig. 2(a), and we hope that it can inherit the land-cover discrimination ability of full-band data.

B. TSR Feature Extraction

HSIs are naturally represented as third-order tensors, with two spatial dimensions and one spectral dimension. Under the assumption of additive Gaussian noise, the observed HSI $\mathcal{Z} \in \mathbb{R}^{h\times v\times s}$ can be modeled as

$$\mathcal{Z} = \mathcal{X} + \mathcal{N} \tag{3}$$

where $\mathcal{X} \in \mathbb{R}^{h\times v\times s}$ denotes the latent clean HSI, $\mathcal{N}$ is zero-mean white Gaussian noise, and h, v, and s represent the height, width, and number of spectral bands, respectively.

Compared to matrix-based methods, tensor-based methods can integrate both spectral and spatial information, enabling a comprehensive capture and processing of the data. Kilmer and Martin [40] proposed a novel tensor decomposition form, referred to as t-SVD, and a tensor–tensor product, referred

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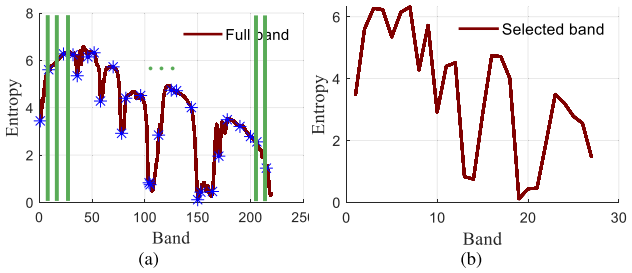


Fig. 2. Feature curves characterized by information entropy. (a) Full band. (b) Select a band group.

to as the tensor product (or t-product). The t-product avoids the inherent information loss caused by unfolding tensors [41]. As a generalization of matrix SVD, t-SVD achieves an optimal Tubal rank, analogous to the matrix rank derived from SVD [42].

Definition 1 (t-Product [40]): Given two third-order tensors $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the t-product $\mathcal{A} * \mathcal{B}$ is a tensor $\mathcal{C} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$

$$\mathcal{C}(i, j, :) = \sum_{k=1}^{n_2} \mathcal{A}(i, k, :) * \mathcal{B}(k, j, :). \tag{4}$$

Theorem 1 (t-SVD): For $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the t-SVD can be written as

$$\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T. \tag{5}$$

In the above equation, $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an f-diagonal tensor, meaning that each of its frontal slices is a diagonal matrix. $\mathcal{U} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ are the orthogonal tensors.

Definition 2 (Tensor Tubal Rank [43]): For $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its tensor tubal rank is denoted as $\text{rank}_t(\mathcal{A})$, and it is defined as follows:

$$\text{rank}_t(\mathcal{A}) = \#\{i, \mathcal{S}(i, i, :) \neq 0\} \tag{6}$$

where $\mathcal{S}$ is obtained from the t-SVD of $\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$. By using t-SVD and t-product, the basis tensor $\mathcal{A}$ can be learned from the permuted tensor $\vec{\mathcal{X}} = \text{permute}(\mathcal{X}, [1, 3, 2])$.

1) Proposed TSR Model: This section focuses on reducing noise interference while extracting discriminative information from the HSI, aiming to improve classification performance. By leveraging the global correlation of HSI in both spatial and spectral domains, a tensor linear representation is employed to analyze LR subspaces, avoiding vectorizing the tensor data. The proposed TSR model for land-cover extraction is as follows:

$$\begin{aligned} &\min_{\mathcal{A},\mathcal{B}} \ \lambda \text{rank}_t(\mathcal{B}) + \frac{1}{2}\|\mathcal{Z} - \mathcal{A} * \mathcal{B}\|_F^2 \\ &\text{s.t.} \quad \mathcal{A}^T * \mathcal{A} = \mathcal{I} \end{aligned} \tag{7}$$

where $\text{rank}_t(\cdot)$ is the tensor tubal rank function, $\lambda > 0$ is a tunable regularization parameter, and the term $\|\mathcal{Z} - \mathcal{A} * \mathcal{B}\|_F^2$ encourages $\mathcal{Z}$ to be close to $\mathcal{A} * \mathcal{B}$. The nonconvex and NP-hard nature of (7) prompts us to consider using the TNN as a replacement for $\text{rank}_t(\cdot)$. Therefore, (7) can be transformed into the following convex optimization problem:

$$\begin{aligned} &\min_{\mathcal{A},\mathcal{B}} \ \lambda\|\mathcal{B}\|_* + \frac{1}{2}\|\mathcal{Z} - \mathcal{A} * \mathcal{B}\|_F^2 \\ &\text{s.t.} \quad \mathcal{A}^T * \mathcal{A} = \mathcal{I} \end{aligned} \tag{8}$$

where the orthogonality constraint on $\mathcal{A}$ promotes distinctiveness between each basis of $\mathcal{A}$, while also helping to reduce noise interference [11]. $\|\cdot\|_*$ is solved using the tensor singular value thresholding algorithm for the TNN of $\mathcal{B}$. Additionally, to obtain an approximation of $\text{rank}_t(\mathcal{B})$, the weighted nuclear norm [44] is introduced into (8), which can be further expressed as follows:

$$\begin{aligned} &\min_{\mathcal{A},\mathcal{B}} \ \lambda\|\mathcal{B}\|_{w,*} + \frac{1}{2}\|\mathcal{Z} - \mathcal{A} * \mathcal{B}\|_F^2 \\ &\text{s.t.} \quad \mathcal{A}^T * \mathcal{A} = \mathcal{I} \end{aligned} \tag{9}$$

where the weighted TNN $\|\mathcal{B}\|_{w,*}$ is the sum of the weighted nuclear norm minimization (WNNM) of the frontal slices of $\hat{\mathcal{B}}$, that is,

$$\|\mathcal{B}\|_{w,*} = \sum_{i=1}^{r} \left\|\hat{\mathcal{B}}(:, :, i)\right\|_{w,*} = \sum_{i=1}^{r} \sum_{j=1}^{\min(h,v)} w(j, j, i)\, \varepsilon(j, j, i) \tag{10}$$

where w represents the weight parameter, and ε denotes the singular values of each frontal slice of $\hat{\mathcal{B}}(:, :, i)$.

2) Optimization Algorithm: Proximal alternating minimization (PAM) is an effective optimization algorithm for solving (9). For different variables, (9) is decomposed into the following subfunctions.

1) $\mathcal{B}$-Subproblem:

$$\mathcal{B}^{t+1} = \arg\min_{\mathcal{B}} \lambda\|\mathcal{B}_i\|_* + \frac{1}{2}\left\|\mathcal{Z} - \mathcal{A}^t * \mathcal{B}\right\|_F^2 + \frac{\alpha}{2}\left\|\mathcal{B} - \mathcal{B}^t\right\|_F^2 \tag{11}$$

where α is the proximal parameter, for (11), based on [10] and [41] and the orthogonality of $\mathcal{A}$, we have the following lemma.

Lemma 1 [40]: For third-order tensors $\mathcal{A}$ and $\mathcal{B}$, if $\mathcal{A}$ is an orthogonal tensor, then $\|\mathcal{A} * \mathcal{B}\|_F = \|\mathcal{B}\|_F$.

Then, (11) can be represented as follows:

$$\begin{aligned} \mathcal{B}^{t+1} &= \arg\min_{\mathcal{B}} \lambda\|\mathcal{B}\|_{w,*} \\ &\quad + \frac{1}{2}\left\|\left(\mathcal{A}^t\right)^T \mathcal{Z} - \mathcal{B}\right\|_F^2 + \frac{\alpha}{2}\left\|\mathcal{B} - \mathcal{B}^t\right\|_F^2 \\ &= \arg\min_{\mathcal{B}} \frac{\lambda}{1+\alpha}\|\mathcal{B}_i\|_{w,*} + \frac{1}{2}\left\|\mathcal{B} - \frac{\alpha\mathcal{B}^t + \left(\mathcal{A}^t\right)^T \mathcal{Z}}{1+\alpha}\right\|_F^2. \end{aligned} \tag{12}$$

Following Wang et al. [45], the optimization problem in (12) can be solved using the weighted tensor singular value thresholding method [44].

2) $\mathcal{A}$-Subproblem:

$$\mathcal{A}^{t+1} = \arg\min_{\mathcal{A}^T * \mathcal{A} = \mathcal{I}} \frac{1}{2}\|\mathcal{Z} - \mathcal{A} * \mathcal{B}\|_F^2 + \frac{\alpha}{2}\left\|\mathcal{A} - \mathcal{A}^t\right\|. \tag{13}$$

For the solution of (13), according to Lin et al. [11], we have $\mathcal{A}^{t+1} = \mathcal{V} * \mathcal{U}^T$, where $\mathcal{U}$ and $\mathcal{V}$ are obtained from the t-SVD of $\mathcal{B}^{t+1} * \vec{\mathcal{Z}}^T + \alpha(\mathcal{A}^t)^T = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$.

After determining the solution approach for the aforementioned subproblems, the initial basis tensor $\mathcal{A}^0$ is obtained from $\mathcal{U}_{pre}(:, 1:r, :)$, where $\mathcal{U}_{pre}$ is derived from the t-SVD of the preprocessed data $\vec{\mathcal{Z}}_{pre} = \mathcal{U}_{pre} * \mathcal{S}_{pre} * \mathcal{V}_{pre}^T$, where r is

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defined as the dimension of the subspace. Here, we determine the value of r using the smaller VD value between $\mathrm{VD_{HFC}}$ and $\mathrm{VD_{NWHFC}}$, denoted as p_2. $\mathcal{B}^0$ is initialized as the subspace projection of the original noisy data $\mathcal{Z}^0$. In the subsequent process, $\mathcal{Z}^{k+1}$ is updated through $\mathcal{Z}^{k+1} = \rho\mathcal{X}^k + (1-\rho)\mathcal{Z}$, where $\mathcal{Z} = \mathcal{Z}^0$ is the original observed image, and $\mathcal{X}^k$ is the denoised image. The iterative regularization parameter ρ is empirically set to a fixed value of 0.6.

C. IEFPNR for HSIC

The diverse spatial distribution of land cover highlights the need to focus on the ability of IEFPNR to capture multiscale information. To this end, the multiscale feature principal components (MSF-PCs) presented in [46] is introduced. MSF-PCs can effectively maintain the main multiscale components of ground objects while reducing the dimension through multiscale spectral extraction and fusion. However, MSF-PCs still has some limitations in the use of spatial information. Its multiple PCAs may lead to the loss of object edge information, thereby causing misclassification of boundary areas. To reduce boundary misclassification and maximize the benefits of the proposed IEFPBS, IEFPNR uses the selected band data $\mathcal{Z}_{\mathbf{B}}^k$ to replace the SPCA features of MSF-PCs.

In addition, inspired by [28], we construct a new data cube using the classification results from the previous iteration. The original spectra are constrained by spatial context information, which allows for the extraction of more class-discriminative features and leads to improved classification performance. Specifically, the spatial distribution of each class in the classification results is treated as pseudo-prior knowledge to construct the enhanced band data cube $\mathcal{Z}_{\mathbf{B}}^k$. With this class-guided spatial constraint, the enhanced feature extraction process can better emphasize interclass discrimination and effectively preserve and strengthen discriminative information.

Algorithm 1 outlines the process for implementing IEFPNR for HSIC, which can be divided into three stages.

In the first stage, IEFP-basedBS [Fig. 1(a)], the target band data $\mathcal{Z}_{\mathbf{B}}^k$ are selected using the proposed IEFPBS.

In the secondstage, feature extraction [Fig. 1(b)], it includes two parts: TSR feature extraction [Fig. 1(b1)]. The proposed TSR model is applied for $\mathcal{Z}_{\mathbf{B}}^k$ to extract land-cover feature $\mathcal{X}_{\mathbf{B}1}^k$; and multiscale spatial feature extraction [46] [Fig. 1(b2)]. Since ground objects exhibit diverse spatial structure and distribution features, MSF extraction is used to capture the supplementary information $\mathcal{X}_{\mathbf{B}_2}^k$ for $\mathcal{X}_{\mathbf{B}_1}^k$. And then, the produced enhance feature $\mathcal{L}^k = \begin{pmatrix} \mathcal{X}_{\mathbf{B}_1}^k \\ \mathcal{X}_{\mathbf{B}_2}^k \end{pmatrix}$ is used for classifying the land cover and obtaining the dmap^k.

In the third stage, class featureguidance [Fig. 1(c)], to achieve stable classification performance, in the kth iteration, dmap^k is used to obtain class feature information, denoted as $\mathcal{D} = \{D_c^k\}_1^C$. The term $D_c^{(k)}$ is defined as

$$D_c^{(k)} = \begin{cases} 1, & \text{if pixel at } \mathcal{Z}(x,y) \in \text{class } c \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

$\mathcal{D}$ provides guiding knowledge about the distribution of different ground objects, which helps suppress interference from intraclass spectral variations and enhances intraclass similarity. It is incorporated as a feedback to update $\mathcal{Z}_{\mathbf{B}}^k$, resulting in $\mathcal{Z}_{\mathbf{B}}^{(k+1)} = \mathcal{Z}_{\mathbf{B}}^k \cup \mathcal{D}$.

Repeat the process and check whether Tanimoto index $\mathrm{TI}^{(k)}$ [7] satisfies the following equation:

$$\mathrm{TI}^{(k)} = \frac{1}{C}\sum_{c=1}^{C} \frac{\left|D_c^{(k-1)} \cap D_c^{(k)}\right|}{\left|D_c^{(k-1)} \cup D_c^{(k)}\right|} > \delta \tag{15}$$

where δ is a given threshold, k denotes the kth iteration, and C represents the number of classes.

By checking whether the $\{D_c^k\}_{c=1}^C$ value of $\mathrm{TI}^{(k)}$ in two consecutive iterations meets the threshold or if the maximum number of iterations K is satisfied, we determine whether to stop the iteration.

Algorithm 1 IEFPNR for HSIC

Input: Observed HSI $\mathcal{Z}$, parameters λ, r.
Initialization: $\mathcal{A}^0$, $\mathcal{B}^0 = (\mathcal{A}^0)^T * \mathcal{Z}$, $\mathcal{X}^0 = \mathcal{O}$, $\partial = 10^{-3}$, $K = 10$, $T = r$.

```
while 0 ≤ k ≤ maxiter and not converged do
    while 0 ≤ t ≤ maxiter and not converged do
        Updating B^t via (12)
        Updating A^t via (13)
        Checking the convergence conditions
            ||A^{t+1} * B^{t+1} − A^t * B^t||_F^2 / ||A^t * B^t|| ≤ ∂
        t = t + 1;
    end while
    Let X_{B_1}^k =inv-Permute(A^{t+1} * B^{t+1})
    Conduct multi-PCA to obtain X_{B_2}^k;
    Construct L^k = (X_{B_1}^k ; X_{B_2}^k);
    Implement SVM on L^k to obtain d;
    Check the iteration stopping conditions:
        TI^{(k)} = (1/C) Σ_{c=1}^{C} |D_c^{(k-1)} ∩ D_c^{(k)}| / |D_c^{(k-1)} ∪ D_c^{(k)}| > δ
    Extract class feature from dmap^k to obtain D;
    Z_B^k = Z_B^k ∪ D
    k = k + 1;
end while
```

Output: Result map

III. Experimental Results and Performance Analysis

A. Experimental Datasets

1) IP[1]: This data was collected by the AVIRIS sensor over the IP test site in northwestern Indiana. It consists of 145×145 pixels across 224 spectral bands. This scene includes regions such as agricultural fields, railroads, and low-density residential areas.

[1]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

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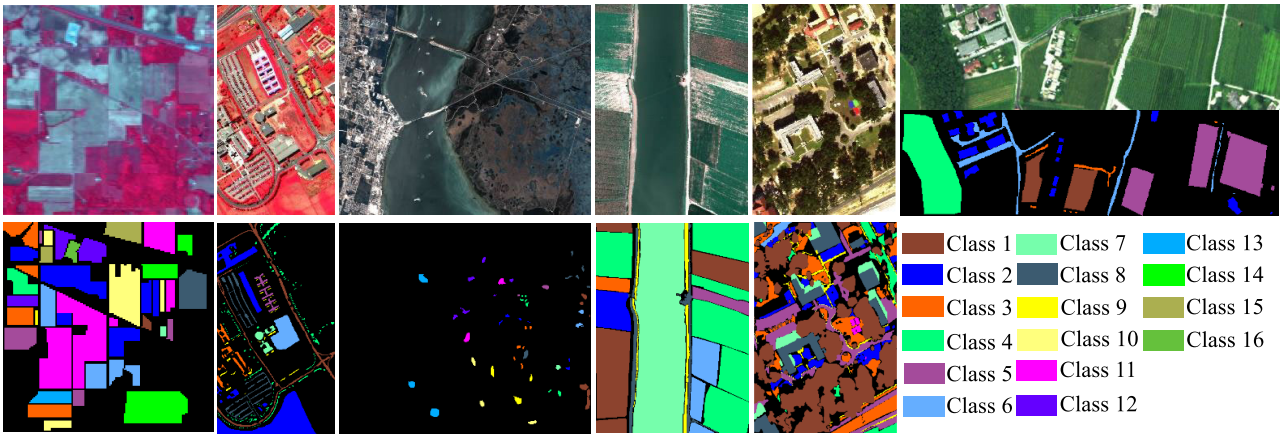


Fig. 3. Three-band false-color images and ground truth (GT) of six scenes.

2) University of Pavia[1]: The University of Pavia (UP) dataset was acquired by the ROSIS sensor and contains labels for nine ground objects. The dataset consists of 610 × 340 pixels with 103 spectral bands. This dataset has been widely used in the HSIC study [22].

3) KSC[1]: This dataset was acquired by the AVIRIS sensor. It consists of 5211 pixels and 176 spectral bands, labeled with 13 classes.

4) Longkou[2]: The WUH-Hi-Longkou dataset was collected in Longkou, Hubei, China, with a resolution of 500 × 400 pixels and 270 spectral bands, labeled with nine classes. The dataset was acquired at an altitude of 500 m, with a spatial resolution of 0.463 m [47], [48].

5) MUUFL[3]: The MUUFL scene was acquired at the University of Southern Mississippi's Gulf Park campus [49]. This scene consists of 325 × 220 pixels, each containing 64 spectral bands, and has been labeled with 11 classes [50], [51].

6) Trento[4]: This dataset was captured in a rural area in southern Italy. It consists of 166 × 600 pixels, with each pixel containing 63 spectral bands in the range of 402.89–989.09 nm.

The false-color images of the six scenes and their corresponding class labels are shown in Fig. 3.

B. Metrics

Several widely used evaluation metrics are employed to assess the effectiveness of different methods, including class accuracy (CA), overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. The final results of all experiments are averaged over 20 independent runs. To facilitate a clearer comparison of the quantitative performance, the best and second-best results are highlighted in bold and underlined, respectively.

C. Comparison Algorithms

In this section, we compare IEFPNR with ten existing algorithms, including feature extraction-based and feature selection-based classification methods: EPF [52], S3-PCA [53], MSF-PCs [46], SpaSSA [54], TensorSSA [55], S3-ULDA [56], LCEM [57], SPCLRSR-DID [58], EPCS-WF [59], and NGNMF-E2DSSA [22], which is a BS and spatial noise removal method. All extracted features by 11 methods are classified using the support vector machine (SVM) implemented via the LIBSVM library, with a Gaussian kernel and fivefold cross-validation [52]. S3-PCA, MSF-PCs, SpaSSA, TensorSSA, S3-ULDA, and LCEM extract features from HSIs using various techniques, including the integration of spatial and spectral information, MSF, and the combination of global and local structural characteristics. NGNMF-E2DSSA introduces a DR method by BS and noise reduction. Other state-of-the-art HSIC methods are also used for comparison. For instance, EPCS-WF uses energy-based, phase-induced Gabor feature selection, aiming to eliminate the effects of redundancy and noise. SPCLRSR-DID considers the global and local structures. And it designed a discriminative incremental dictionary based on preclassification results. We set the parameters for all comparison methods based on the parameter analysis provided in the corresponding articles.

Before the experiments, the n_{BS} values for different datasets are defined as $n_{\mathrm{BS}} = \max\{\mathrm{VD}^{\mathrm{HFC}}, \mathrm{VD}^{\mathrm{NWHFC}}\}$ and summarized in Table I.

TABLE I
VD VALUES OF DIFFERENT DATASETS

	IP	UP	KSC	Trento	MUUFL	Longkou
$\mathrm{VD}^{\mathrm{HFC}}$	27	14	44	10	18	19
$\mathrm{VD}^{\mathrm{NWHFC}}$	18	14	24	10	13	17

D. Results and Analysis

The quantitative performance of different methods across multiple datasets is presented in Tables II–VII. Specifically, the IP dataset shows results with 7% of samples used for classification, and the UP dataset uses 0.1%. For the kennedy space center (KSC), Longkou, MUUFL, and Trento datasets, the experiments are conducted with 9%, 0.5%, 9%, and 0.5% of training samples, respectively. The corresponding

[2]http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm
[3]https://github.com/GatorSense/MUUFLGulfport
[4]https://data.europa.eu/data/datasets/theses-of-the-university-of-trento?locale=en

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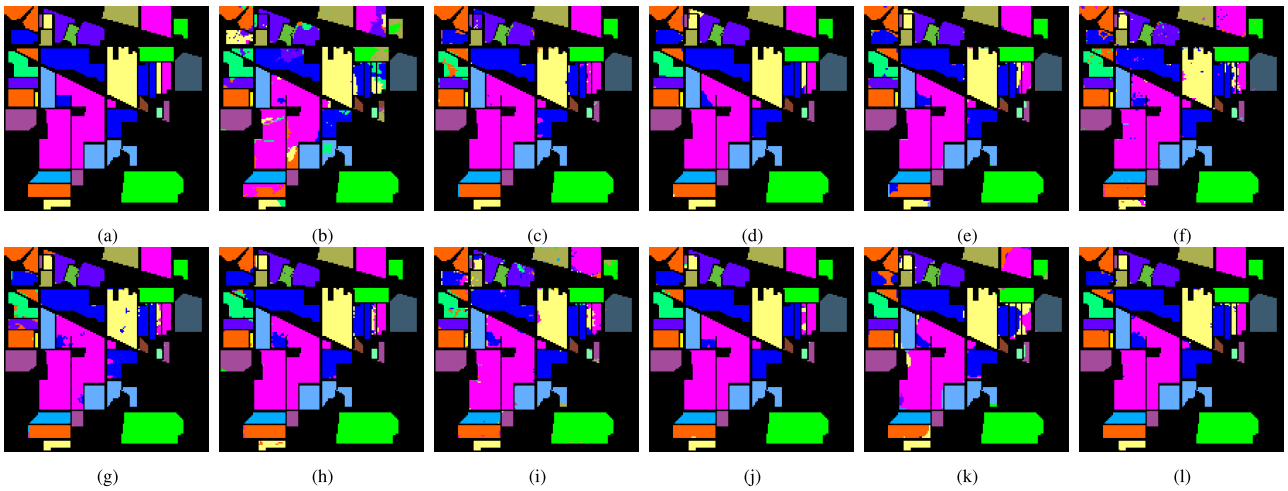


Fig. 4. Classification maps obtained by different methods for IP with 7% labeled samples [OA (%)]. (a) GT(OA). (b) EPF(87.80). (c) S3-PCA(96.02). (d) MSF-PCs(98.79). (e) SpaSSA(97.57). (f) TensorSSA(96.49). (g) S3-ULDA(97.44). (h) LCEM(97.79). (i) NGNMF-E2DSSA(93.97). (j) SPCLRSR-DID(99.12). (k) EPCS-WF(94.88). (l) IEFPNR(99.35).

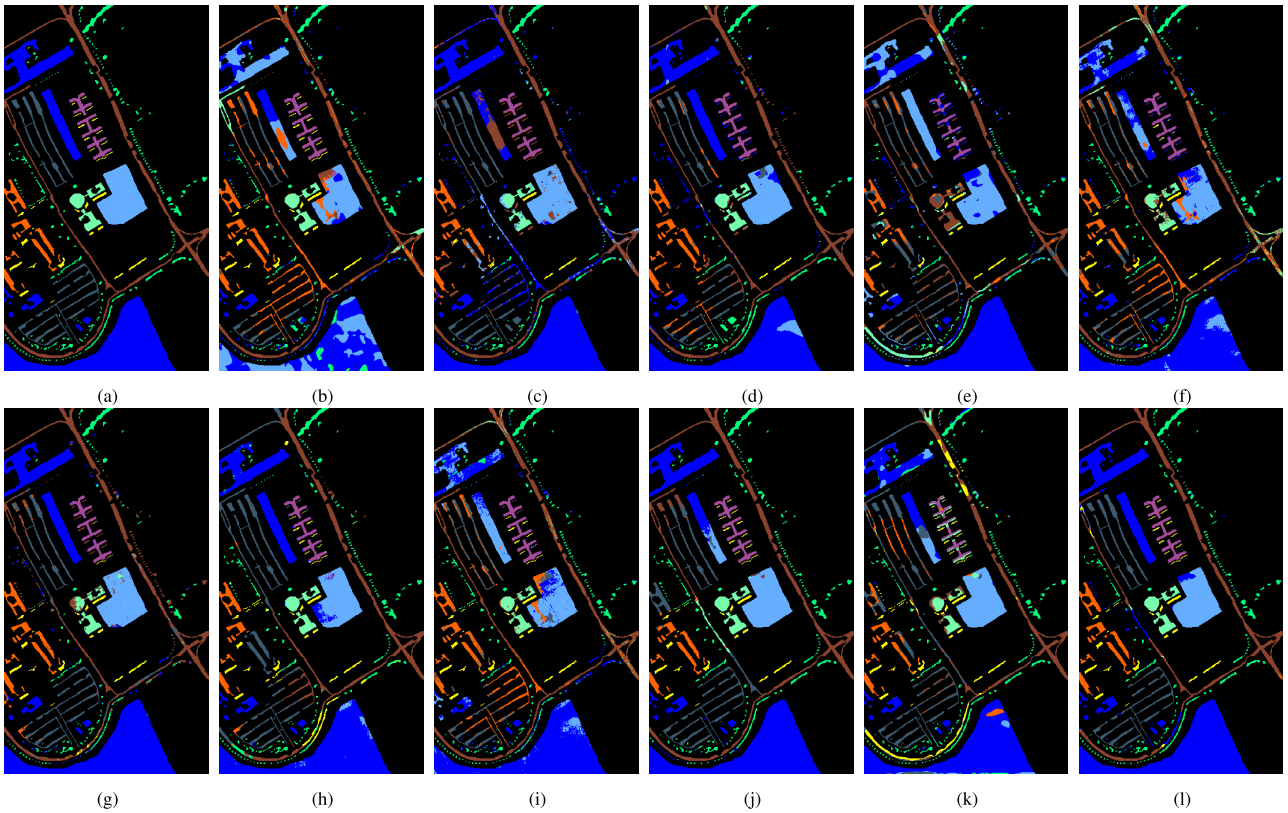


Fig. 5. Classification maps obtained by different methods for UP with 0.1% labeled samples [OA (%)]. (a) GT(OA). (b) EPF(70.28). (c) S3-PCA(84.39). (d) MSF-PCs(89.12). (e) SpaSSA(71.99). (f) TensorSSA(80.38). (g) S3-ULDA(91.35). (h) LCEM(89.72). (i) NGNMF-E2DSSA(78.52). (j) SPCLRSR-DID(93.37). (k) EPCS-WF(79.32). (l) IEFPNR(96.74).

classification maps are shown in Figs. 4–9, which can offer supplementary information to support the quantitative analysis results.

For the IP dataset, as shown in Table II, IEFPNR achieved the highest OA, AA, and Kappa values. In detail, IEFPNR outperformed other methods in 10 out of the 16 land-cover classes. These results highlight IEFPNR's ability to adapt to diverse class features and effectively address class imbalance. This improvement is especially obvious when IEFPNR is compared with NGNMF-E2DSSA, a method that also incorporates BS and noise reduction. Moreover, IEFPNR also outperforms MSF-PCs, which likewise consider MSF. As an important supplement to the quantitative analysis, Fig. 4 provides an intuitive visualization of the classification maps produced by different methods. It can be observed from Fig. 4 that obvious misclassifications still occur in EPF, SpaSSA, LCEM, NGNMF-E2DSSA, and S3-PCA, all of those take spatial–spectral features into account. S3-ULDA enhances S3-

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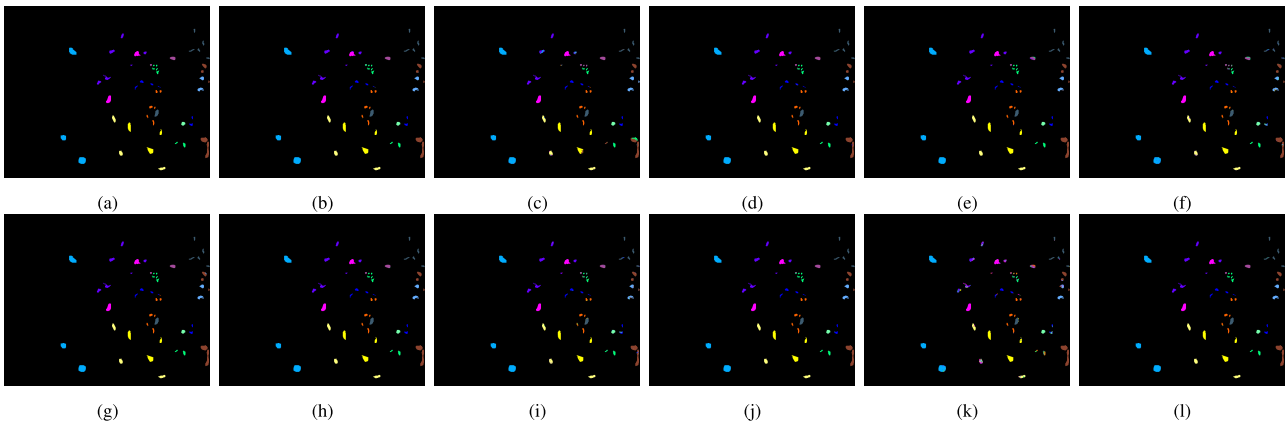


Fig. 6. Classification maps obtained by different methods for KSC with 9% labeled samples [OA (%)]. (a) GT(OA). (b) EPF(99.10). (c) S3-PCA(91.84). (d) MSF-PCs(99.57). (e) SpaSSA(99.42). (f) TensorSSA(95.03). (g) S3-ULDA(99.55). (h) LCEM(99.67). (i) NGNMF-E2DSSA(96.57). (j) SPCLRSR-DID(99.77). (k) EPCS-WF(82.80). (l) IEFPNR(99.71).

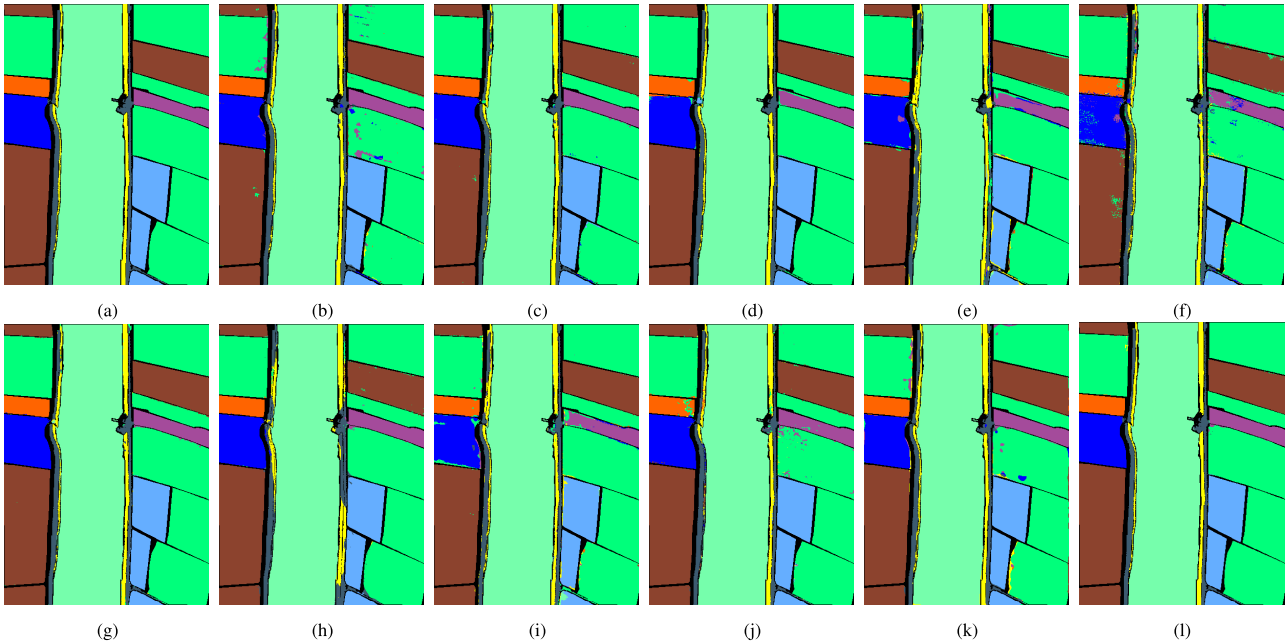


Fig. 7. Classification maps obtained by different methods for Longkou with 0.5% labeled samples [OA (%)]. (a) GT(OA). (b) EPF(98.85). (c) S3-PCA(99.35). (d) MCF-PCs(99.44). (e) SpaSSA(98.60). (f) TensorSSA(97.83). (g) S3-ULDA(99.50). (h) LCEM(98.14). (i) NGNMF-E2DSSA(98.68). (j) SPCLRSR-DID(99.09). (k) EPCS-WF(98.65). (l) IEFPNR(99.79).

PCA by integrating both global and local features. However, misclassifications persist in classes 2, 4, and 10. EPCS-WF shows misclassification in classes 2 and 11. Although TensorSSA considers the 3-D structure of HSI, it still misclassifies pixels in classes 2 and 10. Compared to other comparative methods, SPCLRSR-DID achieves higher accuracy through incremental learning, but there is still room for improvement.

The UP dataset presents a complex background. The classification results for UP dataset are presented in Table III and Fig. 5. As shown, IEFPNR achieves the best performance, with an OA of 96.74%, an AA of 96.38%, and a Kappa of 0.956%, followed by SPCLRSR-DID. The result of IEFPNR, S3-ULDA, S3-PCA, LCEM, TensorSSA, SPCLRSR-DID, and MSF-PCs are visually superior to those of EPF, SpaSSA, NGNMF-E2DSSA, and EPCS-WF, which is consistent with the quantitative results presented in Table III. TensorSSA, S3-PCA, and LCEM have obvious misclassification, particularly in classes 4 and 9, due to their limited use of spatial information. In contrast, the use of MSF in MSF-PCs, as well as the effective use of both global and local spatial features in S3-ULDA, contributes to better differentiation of land-cover classes. However, unsatisfactory classification performance was still observed for classes 4 and 8. Guided by the preclassification results, SPCLRSR-DID achieved relatively good performance, but misclassifications remained in classes 2 and 8. In contrast, IEFPNR effectively alleviated these issues, especially for classes 4 and 8.

Table IV presents the classification result on KSC data under 9% samples. In terms of OA and AA, SPCLRSR-DID achieved the highest performance, with IEFPNR following

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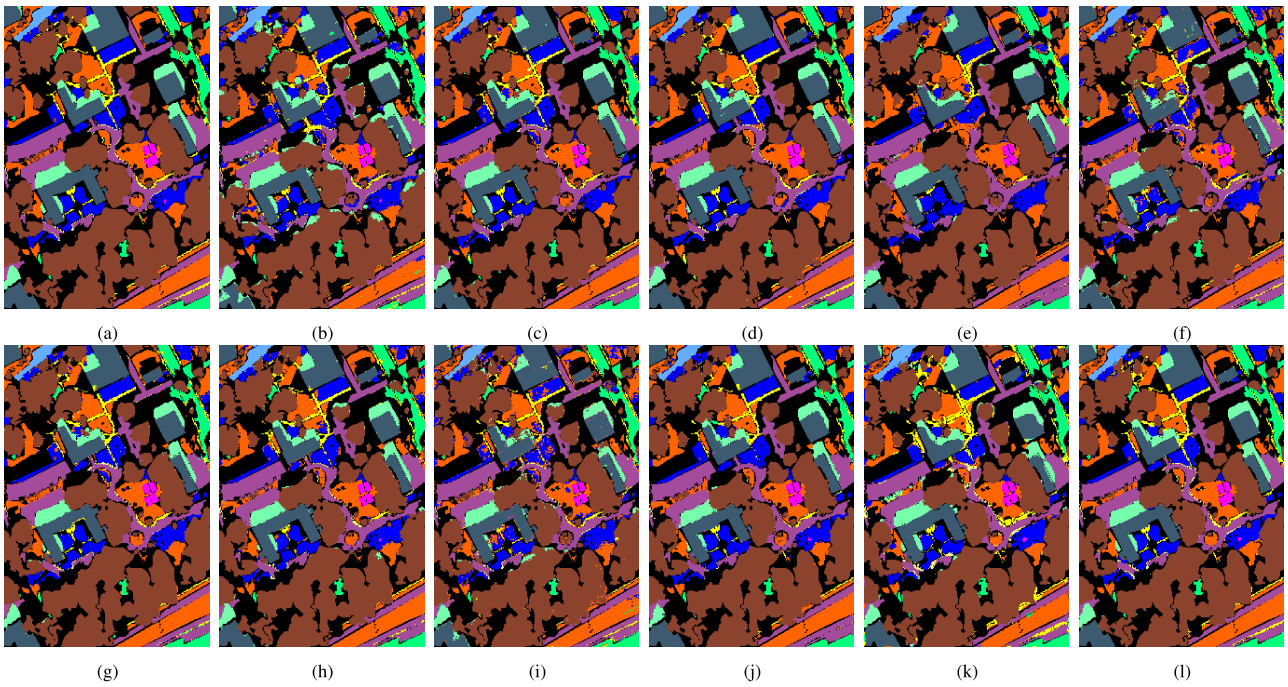


Fig. 8. Classification maps obtained by different methods for MUUFL with 9% labeled samples [OA (%). (a) GT(OA). (b) EPF(90.94). (c) S3-PCA(95.65). (d) MSF-PCs(96.76). (e) SpaSSA(89.27). (f) TensorSSA(92.73). (g) S3-ULDA(95.39). (h) LCEM(94.50). (i) NGNMF(90.42). (j) SPCLRSR-DID(95.40). (k) EPCS-WF(90.16). (l) IEFPNR(98.39).

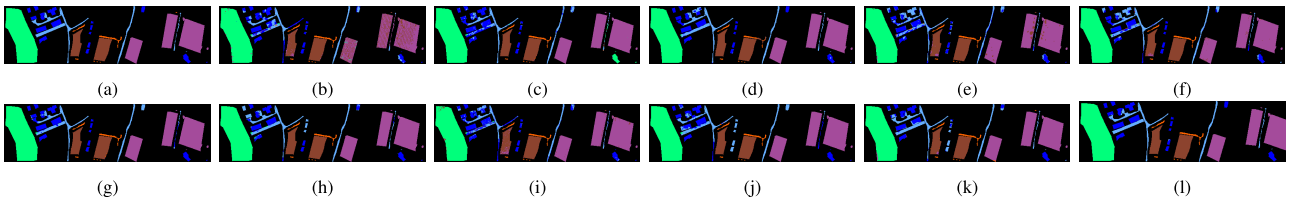


Fig. 9. Classification maps obtained by different methods for Trento with 0.5% labeled samples [OA (%)]. (a) GT(OA). (b) EPF(84.02). (c) S3-PCA(94.72). (d) MCF-PCs(97.71). (e) SpaSSA(92.21). (f) TensorSSA(94.27). (g) S3-ULDA(97.80). (h) LCEM(97.84). (i) NGNMF(93.42). (j) SPCLRSR-DID(96.86). (k) EPCS-WF(97.46). (l) IEFPNR(99.38).

closely behind. In terms of AA, our proposed IEFPNR performed the best, with AA = 99.74%. These results indicate that IEFPNR has a stable capability to distinguish different land-cover classes and produce reliable classification maps. Fig. 6 visually displays the classification maps corresponding to Table IV. As a complement to the quantitative analysis, Fig. 6 provides a visual comparison of the classification results obtained by different algorithms.

For Longkou dataset, the classification results for nine classes obtained by all methods are presented in Fig. 7. Unlike the complex spatial distribution in UP, Longkou has a smoother spatial structure. Therefore, the classification maps in Fig. 7 can distinguish some weaker performing methods. For instance, LCEM shows poor performance in classes 8 and 9. EPF, TensorSSA, EPCS-WF, and SPCLRSR-DID show weaker visual results in classes 4 and 5. NGNMF-E2DSSA and SpaSSA have an obvious misclassification in classes 2 and 5. In contrast, S3-PCA, MSF-PCs, ULDA, and IEFPNR present outstanding visual performance. Table V presents the quantitative evaluation corresponding to Fig. 7. The proposed IEFPNR achieves the highest scores in OA, AA, and Kappa, with particularly good performance in the AA metric, reaching 99.69%. This marks a substantial improvement over the second-best method.

Fig. 8 provides the classification results for 11 labeled classes of the MUUFL dataset using 9% samples. The results show that TensorSSA, LCEM, and NGNMF-E2DSSA have obvious isolated noise points. Among the 11 classes, IEFPNR achieved the best performance in ten classes, significantly outperforming other methods. Furthermore, Table VI shows that all ten comparative algorithms perform poorly on class 10, while IEFPNR achieved a high classification accuracy with $CA_{class10} = 95.82\%$. These result demonstrates the strong robustness of IEFPNR, as it consistently maintains stable performance when adapting to different land-cover classes.

Table VII presents the classification results for the Trento dataset with a sample size of 0.5%. IEFPNR achieved the highest OA, AA, and Kappa values, and outperformed all other methods, representing a substantial improvement over the second-best approach. The corresponding classification map is shown in Fig. 9. MSF-PCs exhibit obvious misclassifications in classes 2 and 3. NGNMF-E2DSSA performed well in class

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TABLE II
CLASSIFICATION ACCURACY (IN %) OF ALL COMPARED METHODS FOR THE IP DATASET WITH 7% TRAINING SAMPLES PER CLASS

Metrics	EPF [52]	S3-PCA [53]	MSF-PCs [46]	SpaSSA [54]	TensorSSA [55]	S3-ULDA [56]	LCEM [57]	NGNMF-E2DSSA [22]	SPCLRSR-DID [58]	EPCS-WF [59]	IEFPNR
1	99.34	100.00	95.23	85.71	97.61	97.61	88.74	95.34	100.00	100.00	100.00
2	71.32	89.90	97.06	96.53	93.22	94.50	88.09	86.35	99.22	89.40	98.66
3	70.62	97.66	98.44	93.38	94.42	97.40	85.62	92.25	99.03	94.30	98.18
4	100.00	80.45	97.72	97.27	96.36	91.36	86.94	55.55	97.19	99.37	100.00
5	94.16	99.33	100.00	97.77	98.66	99.33	86.71	96.94	100.00	96.68	98.94
6	99.39	100.00	100.00	99.85	99.36	100.00	90.72	97.25	100.00	98.50	100.00
7	97.14	96.15	100.00	100.00	96.15	96.15	90.55	88.46	85.51	100.00	100.00
8	100.00	99.77	100.00	99.32	99.32	100.00	90.90	99.55	100.00	99.86	100.00
9	89.00	100.00	100.00	94.44	100.00	100.00	77.27	94.73	100.00	100.00	100.00
10	84.38	85.27	98.00	95.23	91.36	93.90	85.55	87.97	98.90	93.97	98.97
11	83.69	99.95	99.69	98.77	97.06	98.99	90.10	92.15	98.65	89.16	99.45
12	90.48	92.55	94.55	94.01	93.82	92.37	88.43	85.96	99.04	94.50	99.61
13	99.46	98.94	98.94	98.94	97.36	99.47	90.76	97.42	100.00	99.93	99.37
14	95.65	99.91	100.00	100.00	99.57	99.91	90.80	97.91	99.68	98.08	100.00
15	94.45	100.00	100.00	99.72	98.32	99.72	90.90	93.16	99.94	99.46	99.55
16	100.00	98.83	100.00	100.00	100.00	96.51	89.95	90.90	99.65	100.00	100.00
OA	86.58	96.02	98.79	97.57	96.30	97.44	97.69	95.22	98.89	93.85	**99.32**
AA	91.82	96.17	98.72	96.93	97.03	97.32	97.08	93.84	98.77	97.07	**99.54**
Kappa	0.853	0.954	0.986	0.972	0.957	0.970	0.974	0.945	0.990	0.929	**0.992**

TABLE III
CLASSIFICATION ACCURACY (IN %) OF ALL COMPARED METHODS FOR THE UP DATASET WITH 0.1% TRAINING SAMPLES PER CLASS

Metrics	EPF [52]	S3-PCA [53]	MSF-PCs [46]	SpaSSA [54]	TensorSSA [55]	S3-ULDA [56]	LCEM [57]	NGNMF-E2DSSA [22]	SPCLRSR-DID [58]	EPCS-WF [59]	IEFPNR
1	76.20	67.55	94.38	68.25	80.85	86.97	85.17	84.88	97.71	70.80	88.97
2	73.11	92.84	95.46	78.60	82.63	99.56	96.40	80.00	99.28	80.26	99.34
3	49.38	91.45	97.32	49.76	79.81	97.70	71.49	86.35	69.98	75.99	99.57
4	68.47	55.91	63.39	61.17	81.89	49.18	76.67	78.20	96.48	80.08	91.10
5	84.60	95.53	96.57	73.34	97.39	97.24	89.27	98.58	99.18	95.55	99.57
6	42.81	95.60	90.54	85.00	73.68	96.01	93.63	70.43	93.15	83.26	98.34
7	96.04	86.97	95.03	10.54	83.43	76.28	90.00	87.57	76.42	90.25	98.75
8	74.54	81.04	76.78	74.76	66.72	89.96	71.32	52.33	87.69	76.11	97.28
9	98.21	46.30	13.95	56.76	89.64	76.21	53.39	92.11	99.93	78.60	94.47
OA	70.28	84.39	89.12	71.99	80.38	91.35	89.72	78.52	93.37	79.32	**96.74**
AA	73.71	79.24	80.38	62.02	81.78	85.46	83.87	81.17	91.09	81.21	**96.38**
Kappa	0.687	0.791	0.854	0.633	0.746	0.884	0.864	0.723	0.912	0.735	**0.956**

4 by integrating BS and noise reduction. However, it did not achieve satisfactory results in several other classes. In contrast, IEFPNR demonstrated excellent classification accuracy across all classes. These improvements are due to effectively preserving key discriminative information while accounting for the diverse structural features of land-cover classes.

E. Comparison With Deep Learning Classification Methods

In addition to the ten conventional comparison methods mentioned above, this section further includes five recently published deep learning-based HSIC methods for comprehensive performance evaluation: hybrid spectral CNN (HybridSN) [60], weighted feature fusion of CNN and graph attention network (WFCG) [61], spectral–spatial feature tokenization transformer (SSFTT) [62], spectral–spatial score fusion attention network (S3FAN) [63], and hypergraph convolutional network with multiple hyperedge fusion (HGCN-MHF) [64]. To ensure a fair and consistent evaluation, all experiments were conducted on the same hardware platform: an AMD Ryzen 7 5800HS CPU with 16-GB RAM and an NVIDIA GeForce MX450 GPU with CUDA 11.6.

Table VIII summarizes the OA, AA, training time [Tra-t(s)], and test time [Tes-t(s)] achieved by different methods across six datasets. The labeled sample proportions for the IP, UP, KSC, Longkou, MUUFL, and Trento datasets are 7%, 0.1%, 9%, 0.5%, 9%, and 0.5%, respectively. As shown in Table VIII, IEFPNR consistently outperforms all five deep learning models in terms of both OA and AA, under different sizes of training samples. On the IP dataset with 7% training samples, IEFPNR achieves an OA of 99.32%, surpassing the second-best model by more than 2%. Similar trends are observed on UP, KSC, Longkou, MUUFL, and Trento. These results demonstrate that IEFPNR effectively explores spatial–spectral features. It can get discriminative features comparable to those learned by deep neural networks. The superior performance

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TABLE IV

CLASSIFICATION ACCURACY (IN %) OF ALL COMPARED METHODS FOR THE KSC DATASET WITH 9% TRAINING SAMPLES PER CLASS

Metrics	EPF [52]	S3-PCA [53]	MSF-PCs [46]	SpaSSA [54]	TensorSSA [55]	S3-ULDA [56]	LCEM [57]	NGNMF-E2DSSA [22]	SPCLRSR-DID [58]	EPCS-WF [59]	IEFPNR
1	99.56	84.68	99.13	100.00	96.96	98.41	99.59	96.46	100.00	98.12	98.89
2	98.31	97.37	99.09	99.54	88.68	97.73	100.00	91.55	100.00	60.43	100.00
3	98.32	98.27	100.00	98.27	96.12	99.13	98.40	96.63	99.91	98.31	99.54
4	95.07	71.17	96.50	100.00	85.15	100.00	98.38	88.46	96.73	52.63	99.16
5	97.76	98.63	97.94	85.61	88.35	100.00	97.39	85.23	100.00	49.52	99.60
6	99.65	100.00	100.00	100.00	89.90	98.55	99.66	84.43	99.90	88.49	99.94
7	100.00	100.00	100.00	100.00	94.73	100.00	100.00	90.72	100.00	98.40	100.00
8	99.83	98.97	99.74	99.74	91.07	100.00	99.71	94.25	99.84	92.50	100.00
9	99.92	90.90	100.00	100.00	99.36	100.00	99.85	100.00	100.00	96.92	99.91
10	99.57	90.19	100.00	100.00	90.73	100.00	99.97	94.13	99.73	63.15	99.83
11	99.59	99.73	100.00	100.00	99.73	100.00	100.00	96.14	99.94	91.59	100.00
12	97.37	76.14	100.00	100.00	97.37	100.00	99.97	95.71	100.00	42.89	99.78
13	99.95	98.81	100.00	100.00	97.86	100.00	100.00	100.00	99.97	94.52	100.00
OA	99.10	91.84	99.57	99.42	95.03	99.55	99.67	96.57	**99.77**	82.80	99.71
AA	98.84	92.71	99.41	98.70	93.54	99.52	99.48	95.40	99.69	79.04	**99.74**
Kappa	0.990	0.909	0.995	0.993	0.944	0.995	0.996	0.961	**0.997**	0.807	0.996

TABLE V

CLASSIFICATION ACCURACY (IN %) OF ALL COMPARED METHODS FOR LONGKOU DATASET WITH 0.5% TRAINING SAMPLES PER CLASS

Metrics	EPF [52]	S3-PCA [53]	MSF-PCs [46]	SpaSSA [54]	TensorSSA [55]	S3-ULDA [56]	LCEM [57]	NGNMF-E2DSSA [22]	SPCLRSR-DID [58]	EPCS-WF [59]	IEFPNR
1	99.56	99.81	99.86	99.03	98.14	99.94	99.68	99.64	99.57	99.26	99.79
2	99.17	99.89	97.13	97.05	92.96	99.90	98.83	94.97	98.92	97.70	99.70
3	97.74	99.46	98.01	96.08	90.41	99.43	98.96	90.61	96.54	99.42	99.97
4	97.85	99.31	99.85	99.03	98.58	99.58	99.30	99.03	99.56	97.88	99.63
5	99.06	99.54	94.16	88.54	90.38	99.63	98.36	87.16	89.30	99.22	99.87
6	99.90	99.93	99.83	97.07	98.35	99.93	98.57	96.68	99.58	99.60	99.98
7	99.97	99.96	99.98	99.96	99.96	99.98	99.93	99.97	99.97	99.46	99.95
8	94.68	96.23	96.27	73.44	88.95	95.59	74.96	88.09	92.63	95.57	98.81
9	95.26	90.94	97.15	84.44	88.35	93.11	74.25	91.90	97.92	96.13	99.42
OA	98.85	99.35	99.44	98.60	97.83	99.50	98.14	98.68	99.09	98.65	**99.79**
AA	98.13	98.34	98.03	95.34	94.01	98.57	94.10	95.78	97.11	98.25	**99.69**
Kappa	0.985	0.991	0.992	0.981	0.971	0.993	0.975	0.982	0.988	0.982	**0.997**

TABLE VI

CLASSIFICATION ACCURACY (IN %) OF ALL COMPARED METHODS FOR THE MUUFL DATASET WITH 9% TRAINING SAMPLES PER CLASS

Metrics	EPF [52]	S3-PCA [53]	MSF-PCs [46]	SpaSSA [54]	TensorSSA [55]	S3-ULDA [56]	LCEM [57]	NGNMF-E2DSSA [22]	SPCLRSR-DID [58]	EPCS-WF [59]	IEFPNR
1	93.24	98.47	98.26	95.57	97.10	98.29	97.13	95.59	98.51	91.81	98.36
2	89.09	93.51	95.80	81.23	84.94	92.43	91.89	80.12	87.42	87.72	98.10
3	84.91	93.59	95.94	86.13	89.61	93.06	91.75	86.25	93.73	83.19	98.43
4	97.39	94.76	95.78	85.79	95.06	94.16	95.11	90.36	94.02	95.55	99.46
5	93.92	95.46	96.63	90.43	93.95	95.20	96.06	93.29	92.04	90.15	97.70
6	98.94	93.63	90.56	91.50	88.44	95.28	94.74	80.89	97.87	99.25	99.82
7	94.31	94.14	94.14	75.09	78.00	93.40	87.49	74.01	96.56	91.52	98.61
8	93.37	96.37	97.79	95.43	94.82	96.65	96.66	91.86	97.11	93.97	99.08
9	52.81	74.04	85.23	30.00	72.69	73.73	71.04	69.12	89.23	75.73	98.33
10	30.87	52.40	79.51	00.00	40.96	50.60	49.69	64.45	47.11	62.13	95.82
11	95.98	89.75	92.21	84.42	91.80	93.85	95.86	90.16	89.37	97.28	98.94
OA	90.94	95.65	96.76	89.27	92.73	95.39	94.50	90.42	95.40	90.16	**98.39**
AA	84.08	88.74	92.90	74.15	84.31	88.78	87.89	83.28	89.36	88.03	**98.42**
Kappa	0.894	0.942	0.957	0.857	0.903	0.939	0.927	0.873	0.939	0.868	**0.978**

of the proposed method is attributed to the well-designed BS strategy of IEFPBS. The bands selected by IEFPBS are jointly determined by VD value and IEF curvature, enabling the proposed method to select discriminative bands that are useful for classification tasks, thereby effectively constraining the feature space for subsequent classification. Furthermore, the designed BS and TSR modeling can remove redundancy and noise while retaining the discriminative information for clas-

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TABLE VII
CLASSIFICATION ACCURACY (IN %) OF ALL COMPARED METHODS FOR THE TRENTO DATASET WITH 0.5% TRAINING SAMPLES PER CLASS

Metrics	EPF [52]	S3-PCA [53]	MSF-PCs [46]	SpaSSA [54]	TensorSSA [55]	S3-ULDA [56]	LCEM [57]	NGNMF-E2DSSA [22]	SPCLRSR-DID [58]	EPCS-WF [59]	IEFPNR
1	97.60	99.62	100.00	96.63	97.83	99.62	99.96	94.91	99.76	98.22	99.97
2	82.73	76.31	90.40	68.38	72.78	92.41	89.16	82.34	83.14	91.02	99.19
3	95.88	92.43	68.69	93.69	94.32	96.00	97.26	92.85	97.19	98.61	100.00
4	97.95	100.00	100.00	99.46	99.91	100.00	99.98	99.57	99.29	99.66	100.00
5	70.06	99.98	100.00	95.99	98.78	100.00	99.86	98.83	99.90	99.28	99.47
6	72.26	73.11	91.70	74.85	78.30	87.11	90.32	66.30	89.49	89.80	96.67
OA	84.02	94.72	97.71	92.21	94.27	97.80	97.84	93.42	96.86	97.46	**99.38**
AA	86.08	90.24	91.80	88.17	90.32	95.86	96.09	89.13	94.80	96.10	**99.21**
Kappa	0.827	0.929	0.969	0.896	0.923	0.970	0.971	0.912	0.958	0.966	**0.991**

TABLE VIII
COMPARISON OF CLASSIFICATION ACCURACY (IN %) OF THE PROPOSED METHOD WITH DEEP LEARNING METHODS

Dataset	Metrics	HybridSN [60]	SSFTT [62]	WFCG [61]	S3FAN [63]	HGCN-MHF [64]	IEFPNR
IP	OA	93.83	98.03	91.75	94.63	97.31	**99.32**
	AA	87.12	96.33	91.97	97.21	98.68	**99.54**
	Tra-t(s)	7.48×10^0(G)	8.66×10^1(G)	1.02×10^2(G)	2.17×10^1(G)	1.71×10^2(G)	-
	Tes t(s)	2.05×10^0(G)	3.58×10^0(G)	0.46×10^1(G)	8.80×10^0(G)	$0.05 \times 10^{(-1)}$(G)	3.73×10^1(C)
UP	OA	76.45	76.96	90.15	73.90	91.67	**96.74**
	AA	52.97	54.66	85.80	83.78	94.89	**96.38**
	Tra-t(s)	9.07×10^0(G)	2.98×10^1(G)	1.05×10^2(G)	3.81×10^1(G)	6.97×10^3(G)	-
	Tes-t(s)	5.83×10^0(G)	1.60×10^1(G)	0.46×10^0(G)	6.22×10^1(G)	4.20×10^0(G)	2.71×10^2(C)
KSC	OA	63.83	68.54	90.65	87.73	98.92	**99.71**
	AA	50.64	53.58	90.15	81.23	98.60	**99.74**
	Tra-t(s)	6.21×10^0(G)	6.34×10^1(G)	1.85×10^2(G)	8.50×10^2(G)	9.93×10^3(G)	-
	Tes-t	1.03×10^0(G)	1.48×10^0(G)	0.70×10^0(G)	4.09×10^1(G)	4.61×10^0(G)	2.73×10^3(C)
Longkou	OA	97.96	97.63	89.19	99.09	99.24	**99.79**
	AA	93.47	92.41	91.51	99.00	99.31	**99.69**
	Tra-t(s)	2.44×10^1(G)	8.22×10^1(G)	9.60×10^1(G)	7.90×10^2(G)	1.57×10^3(G)	-
	Tes-t(s)	4.14×10^1(G)	5.11×10^1(G)	0.52×10^0(G)	4.33×10^2(G)	0.15×10^0(G)	4.16×10^2(C)
MUUFL	OA	93.11	93.13	85.80	94.03	95.50	**98.39**
	AA	85.45	79.17	76.40	95.71	95.41	**98.42**
	Tra-t(s)	1.05×10^1(G)	4.18×10^2(G)	1.06×10^2(G)	2.35×10^3(G)	6.72×10^2(G)	-
	Tes-t(s)	7.43×10^1(G)	1.80×10^1(G)	0.51×10^0(G)	8.27×10^1(G)	0.01×10^0(G)	1.13×10^2(C)
Trento	OA	92.87	97.21	89.22	97.30	97.67	**99.38**
	AA	76.12	86.60	88.98	95.49	95.95	**99.21**
	Tra-t(s)	7.83×10^1(G)	4.23×10^2(G)	9.84×10^1(G)	2.72×10^2(G)	1.38×10^3(G)	-
	Tes-t(s)	4.28×10^1(G)	1.10×10^1(G)	0.52×10^0(G)	1.89×10^2(G)	0.03×10^0(G)	8.40×10^1(C)

sification, which is particularly useful for complex scenarios with high spectral redundancy. Moreover, the iterative optimization process guided by class information further enhances intraclass consistency, reduces isolated noise, and makes the classification results more stable and reliable. Due to the above design, IEFPNR achieves outperformance without requiring large-scale labeled data or complex network architectures, which are typically essential for deep learning methods.

To provide a clearer analysis, we have presented the comparison with the time costs of different methods during both training and testing phases, as well as their corresponding hardware requirements. Here, "C" denotes CPU and "G" represents GPU. As shown by the results, although deep learning methods such as HGCN-MHF and S3FAN achieve faster inference speeds, their training processes rely on GPU acceleration, and their overall time consumption is still higher than that of IEFPNR. In contrast, while IEFPNR does not achieve the shortest runtime, its entire classification process can be completed only on a CPU, without the need for high-end hardware, and its computational efficiency remains within an acceptable range. Taking into account the total computational cost, classification accuracy, and hardware deployment requirements, our proposed method has a favorable balance in terms of overall computational burden. This makes IEFPNR particularly suitable for practical applications where computing resources or labeled samples are limited.

F. Evaluation With Different Classifiers

Since the features extracted by seven methods: S3-PCA, MSF-PCs, SpaSSA, TensorSSA, S3-ULDA, NGNMF-E2DSSA, and EPCS-W can also be combined with other classifiers, we further conducted an experiment using ran-

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TABLE IX
CLASSIFICATION PERFORMANCE ON DIFFERENT DATASETS WITH THE KNN CLASSIFIER

Dataset	Metrics	Raw	S3-PCA [53]	MSF-PCs [46]	SpaSSA [54]	TensorSSA [55]	S3-ULDA [56]	NGNMF-E2DSSA [22]	EPCS-WF [59]	IEFPNR
IP	OA	72.98	90.29	97.30	86.86	86.43	97.58	85.26	86.46	**99.23**
	AA	68.89	88.32	97.49	82.81	86.72	96.24	84.19	81.91	**98.25**
UP	OA	65.81	80.77	77.64	68.13	68.74	85.49	67.69	74.98	**88.38**
	AA	53.72	67.70	63.4	53.07	55.94	68.59	55.91	58.76	**70.88**
KSC	OA	82.92	98.01	97.11	95.11	88.55	99.55	91.32	89.41	**99.78**
	AA	76.39	98.1	96.06	92.63	83.40	99.37	86.54	85.37	**99.50**
Longkou	OA	89.40	98.54	96.62	94.91	94.44	98.97	94.14	95.62	**99.54**
	AA	69.70	96.39	88.24	82.63	82.10	97.36	80.61	86.18	**98.45**
MUUFL	OA	84.01	93.48	92.47	84.11	89.59	95.99	88.39	82.68	**98.27**
	AA	74.42	85.59	86.39	70.50	79.72	89.25	78.43	69.35	**94.05**
Trento	OA	78.97	94.36	95.52	94.19	91.56	97.15	93.53	94.78	**98.14**
	AA	77.68	90.01	89.25	82.89	87.37	93.43	88.42	86.33	**95.29**

TABLE X
CLASSIFICATION PERFORMANCE ON DIFFERENT DATASETS WITH THE RF CLASSIFIER

Dataset	Metrics	Raw	S3-PCA [53]	MSF-PCs [46]	SpaSSA [54]	TensorSSA [55]	S3-ULDA [56]	NGNMF-E2DSSA [22]	EPCS-WF [59]	IEFPNR
IP	OA	73.65	97.42	97.83	92.17	87.90	96.92	83.23	92.2	**99.04**
	AA	58.75	**96.90**	94.09	83.24	78.45	94.7	75.95	79.68	92.39
UP	OA	65.76	75.68	81.47	65.60	69.00	80.92	68.97	81.41	**90.28**
	AA	44.25	51.19	61.92	42.40	46.99	56.14	49.39	62.54	**69.31**
KSC	OA	86.28	98.05	99.16	97.15	92.38	99.26	93.67	98.47	**99.64**
	AA	80.27	97.97	98.51	95.58	89.62	98.80	90.31	97.00	**99.53**
Longkou	OA	91.73	99.16	98.82	95.01	95.40	99.16	94.04	96.72	**99.48**
	AA	72.38	97.99	96.90	83.34	83.41	97.40	77.96	88.84	**98.54**
MUUFL	OA	87.10	95.97	96.42	85.89	90.45	95.53	89.82	91.66	**98.74**
	AA	75.37	87.71	88.92	69.78	79.22	86.16	77.91	78.49	**94.01**
Trento	OA	82.70	95.77	96.41	92.99	92.19	96.01	93.52	95.58	**96.86**
	AA	77.95	87.99	86.25	83.43	86.88	88.98	86.97	88.95	**90.02**

dom forest (RF) and k-nearest neighbors (KNN) (with k = 3) classifiers. The labeled sample proportions for the IP, UP, KSC, Longkou, MUUFL, and Trento datasets are 7%, 0.1%, 9%, 0.5%, 9%, and 0.5%, respectively. The results are presented in Tables IX and X. "Raw" represents the classification results from the raw HSI. Consistent with the results obtained using SVM, the proposed method continues to demonstrate excellent performance. For KNN, the results show that across the six datasets, all feature extraction methods improved classification accuracy compared with the raw data, with IEFPNR providing a clear advantage over other methods. For RF, IEFPNR also performed outstandingly. Specifically, it achieved the best OA and AA values on UP, KSC, MUUFL, Longkou, and Trento datasets. On the IP dataset, IEFPNR achieved remarkable OA but did not obtain the highest AA. Nevertheless, it still exhibits strong competitiveness.

In conclusion, the outstanding performance of IEFPNR across different classifiers highlights its effectiveness in preserving discriminative features and resisting interference in HSI. The experimental results further confirm the robustness and effectiveness of the proposed method in HSIC.

IV. DISCUSSION

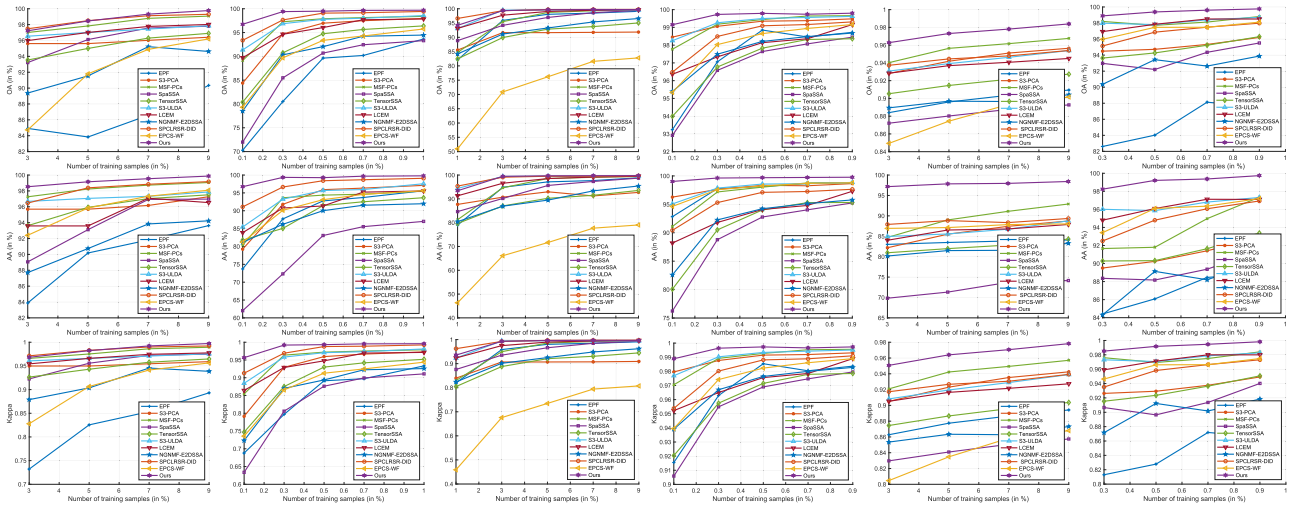
A. Generalization Performance

Experiments with different labeled sample ratios are useful for assessing the performance of the proposed method. Fig. 10 presents the OA, AA, and Kappa value trends of different methods for six datasets. The labeled sample proportions for the IP, UP, KSC, Longkou, MUUFL, and Trento datasets are {3%, 5%, 7%, 9%}, {0.1%, 0.3%, 0.5%, 0.7%, 1%}, {1%, 3%, 5%, 7%, 9%}, {0.1%, 0.3%, 0.5%, 0.7%, 0.9%}, {3%, 5%, 7%, 9%, 11%}, and {0.3%, 0.5%, 0.7%, 0.9%}, respectively. As shown in Fig. 10, the classification accuracy of all methods improves as the number of labeled samples increases. The proposed method has superior performance across all three metrics in most cases for the IP, UP, Longkou, MUUFL, and Trento datasets. Notably, on the UP dataset with a complex background, it significantly outperforms the comparison methods. These results demonstrate the robustness and practical applicability of the proposed method.

B. Ablation Study

To assess the contribution of each component in the proposed method, Table XI presents the results of an ablation

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(a) (b) (c) (d) (e) (f)

Fig. 10. Variation of OA, AA, and kappa values for different methods with different percentages of labeled samples. (a) IP. (b) UP. (c) KSC. (d) Longkou. (e) MUUFL. (f) Trento.

TABLE XI
ABLATION EXPERIMENTAL RESULTS

Dataset	Metrics	A	B	C	D
IP(5%)	OA	95.44	96.90	89.81	**98.46**
	AA	97.03	97.94	94.19	**99.15**
	Kappa	0.947	0.964	0.883	**0.982**
UP(0.1%)	OA	94.34	91.70	73.01	**96.74**
	AA	93.03	89.81	78.94	**96.38**
	Kappa	0.924	0.891	0.650	**0.956**
Longkou(0.5%)	OA	99.38	99.41	99.10	**99.79**
	AA	99.25	99.09	99.00	**99.69**
	Kappa	0.991	0.992	0. 988	**0.997**

study on three datasets as examples. Specifically, column A represents the exclusion of the IEFPBS module, highlighting the positive impact of IEFPBS in the classification result. Column B shows the experimental outcomes when the proposed TSR module is deleted. Column C shows the results without the MSF compensation module. Finally, column D corresponds to the proposed IEFPNR.

As shown in column D in Table XI, IEFPNR achieved the highest value in OA, AA, and Kappa. The result in column B shows that after removing the TSR module, the classification accuracy dropped to OA = 96.90% and an AA = 97.94%, with the Kappa value decreasing from 0.982 to 0.964. The metrics for the UP and Longkou datasets also appear decline, with the significant drop seen in the UP dataset, where OA, AA, and Kappa decreased by 5.04%, 6.57%, and 0.655, respectively. These results demonstrate the positive contribution of the proposed TSR module. Comparing columns C and D further shows that MSF compensation improves the classification metrics. Specifically, OA increased by 8.65%, 23.73%, and 0.69%, while AA improved by 4.96%, 17.44%, and 0.69%. Kappa values rose by 0.099, 0.306, and 0.009, respectively. These improvements demonstrate the effectiveness of the added MSF compensation strategy. Furthermore, the results in column A show that, after removing the IEFPBS module, classification accuracy on the IP dataset declined by 3.02%, 2.12%, and 0.035 for OA, AA, and Kappa, respectively. On the UP dataset, performance decreased by 2.4%, 3.35%, and 0.032, respectively. Similarly, for Longkou dataset, the OA, AA, and Kappa values decreased by 0.41%, 0.44%, and 0.006, respectively. These experimental results emphasize the important role of each module of IEFPNR in improving classification accuracy.

C. Parameter Analysis

The main parameters involved in the proposed method are the regularization parameter λ, the proximal parameter α, and the initial subspace dimension r. To verify the stability and effectiveness of IEFPNR, this section conducts experiments under different parameters. Among them, r is the subspace dimension parameter, which is related to the number of unique spectral signal sources in HSI. Fortunately, VD value is defined as the minimum number of spectral unique signal sources required to characterize HSI from the perspective of target detection and classification [28]. Therefore, we set r to $r = min\{\mathrm{VD}^{\mathrm{HFC}}, \mathrm{VD}^{\mathrm{NWHFC}}\}$ in all experiments. Then, the sensitivity and stability of parameters λ and α are analyzed on different datasets within a reasonable range of values.

According to (9), (11), and (13), it can be seen that the parameter λ controls the strength of noise suppression and regularization. A larger value of λ can strengthen the noise suppression effect, but may over-smooth discriminative spectral details. And a smaller value of λ can retain more information, but may introduce residual noise. The parameter α balances different components in the optimization objective and affects the convergence behavior as well as the tradeoff between stability and discriminability. Fig. 11(a) and (b) shows the changes in OA and AA values of IEFPNR for different datasets when α increases from 0.3 to 0.8. Fig. 11(c) and (d) shows the changes in OA and AA values of IEFPNR when λ takes values in the range of {0.001, 0.002, 0.003,

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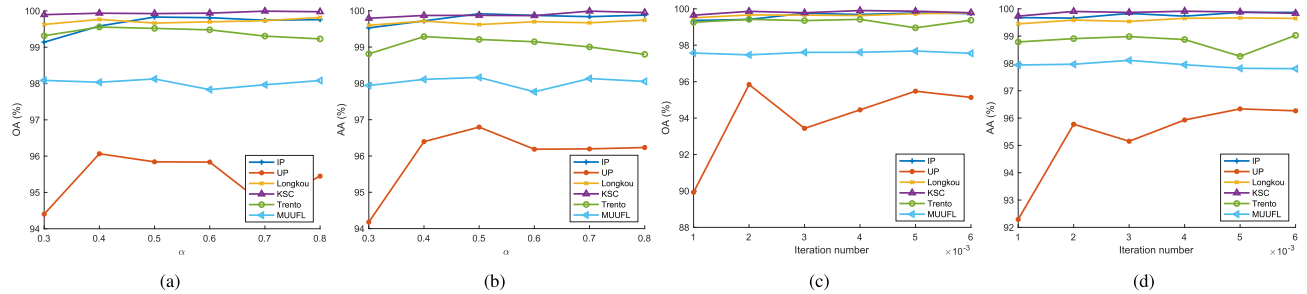


Fig. 11. Sensitivity analysis of the parameters on different datasets. (a) OA versus proximal parameter α. (b) AA versus proximal parameter α. (c) OA versus proximal parameter λ. (d) AA versus proximal parameter λ.

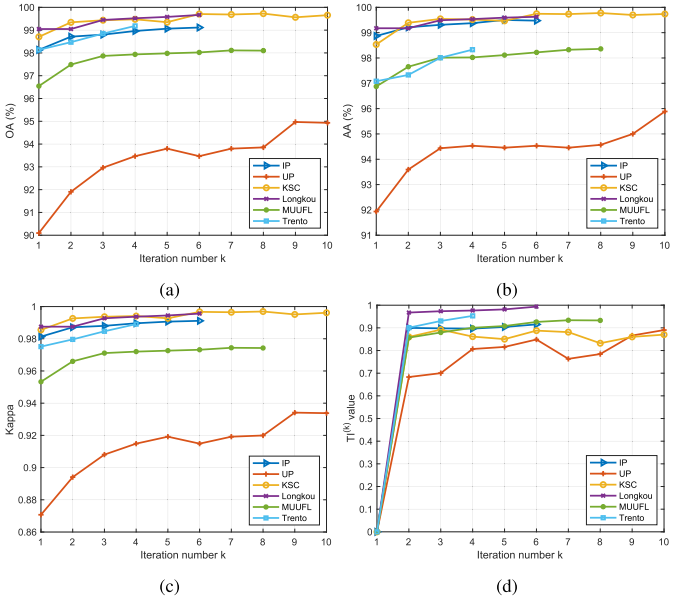


Fig. 12. Curve of classification accuracies versus the iteration number k. (a) OA value versus the iteration number k. (b) AA value versus the iteration number k. (c) Kappa coefficient value versus the iteration number k. (d) $\mathrm{TI}^{(k)}$ value versus the iteration number k.

0.004, 0.005, 0.006}. As shown in Fig. 11, the proposed method maintains stable performance across a wide range of parameter values. Specifically, as the values of α and λ increase, IEFNR maintains stable classification results and achieves high accuracy on the IP, KSC, LongKou, MUUFL, and Trento datasets, while exhibiting slight fluctuations on the UP dataset. Based on the analysis, we empirically set $\alpha = 0.5$ and $\lambda = 0.005$, which lie within the stable region, for all experiments. These parameter analyses also provide guidance for users to adjust the parameters under different HSI scenarios.

D. Iterative Mechanism Analysis

This section presents a comprehensive analysis of the effectiveness of the iterative mechanism in IEFPNR, where the iterative process is terminated based on the maximum iteration number K and the TI index threshold δ. Given the flat land-cover distribution and large homogeneous regions in Longkou dataset, its TI threshold δ was set to 0.99 to prevent premature convergence. For other datasets, a lower threshold of 0.93 was adopted. The corresponding experimental results are presented in Fig. 12. As shown in Fig. 12(d), the $\mathrm{TI}^{(k)}$ values increase significantly during the initial iterations, and subsequently stabilize, reflecting the gradual consistency of the feedback features and the progressive convergence of the iteration process. Correspondingly, the trends of the OA, AA, and Kappa coefficient curves in Fig. 12(a)–(c) are closely consistent with the TI curve, and all metrics improve progressively as the number of iterations increases. Specifically, all metrics exhibit an increasing trend over the first three iterations. Beyond four iterations, the rate of improvement decelerates. This behavior demonstrates that the proposed class feature guidance mechanism can effectively improve classification performance. In addition, the observation shows that the TI index can adaptively terminate the iterations across different datasets, achieving an effective tradeoff between classification performance and computational efficiency.

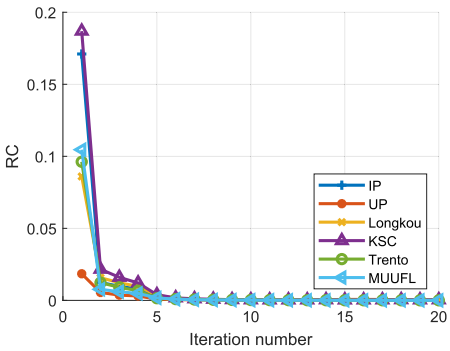


Fig. 13. Changes in RC values with the number of iterations on different datasets.

E. Numerical Convergence Analysis

We demonstrate the convergence of solving the TSR model (9) using the PAM algorithm through numerical analysis. Specifically, we define the relative change (RC) of the tth iteration in (16) and plot the RC curve in Fig. 13. It is evident that as the number of iterations increases, the RC value rapidly approaches 0

$$RC = \frac{\left\| \mathcal{A}^t * \mathcal{B}^t - \mathcal{A}^{t-1} * \mathcal{B}^{t-1} \right\|_F}{\left\| \mathcal{A}^{t-1} * \mathcal{B}^{t-1} \right\|_F}. \tag{16}$$

V. Conclusion

This article presents a novel DR method, namely, IEFPNR. The IEFPNR effectively extracts interpretable low-dimensional

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features in HSI. Specifically, by integrating concepts of information theory and curvature, IEFPBS effectively preserves the discriminative ability of spectral features while mitigating the impact of spectral variability. Then, the TSR model can separate the interference by further extracting land-cover features from the selected band subset. In addition, by considering the various sizes and shapes of ground objects, multiscale features provide valuable supplementary information. These processes provide a more comprehensive and robust representation of land cover. Finally, the iterative refinement process, guided by class features, reduces intraclass spectral variability while enhancing within-class similarity. The experiments on HSI datasets demonstrate that the proposed IEFPNR achieves superior classification results for different datasets.

Although the IEFPNR has outperformed in HSIC, there are still some limitations. For example, some parameters still need to be manually set, which limits the adaptability of IEFPNR. Future research will explore multisource data fusion and automated parameter adjustment mechanisms to enhance the practicality and generalization capability.

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